

Exercise 3: Normal Forms

Introduction to Database Systems – Spring 2025

Martin Hentschel and Omar Shahbaz Khan

In both parts of this exercise, functional dependency refers to full functional dependency.

1. Analog Database Schema

We thank the Analog app's development team for allowing us to use their database schema in class!

On page 3 of this exercise, you are given the original database schema of the backend that powers the Analog app. Even for a relatively small app, the database schema can consist of many tables. In this exercise, we will focus on the following three tables/relations:

Users(Id, Email, Name, Password, Salt, Experience, DateCreated, DateUpdated, IsVerified, PrivacyActivated, UserGroup, UserState, Programme_Id)

Products(Id, Price, NumberOfTickets, Name, Description, ExperienceWorth, Visible)

Purchases(Id, ProductName, ProductId, Price, NumberOfTickets, DateCreated, OrderId, ExternalTransactionId, Status, PurchasedBy_Id, Type)

Solve the following tasks:

- a) Identify all functional dependencies. Assume that users can sign up multiple times with the same email address and user name, and that multiple products can have the same name and description.
- b) Given these functional dependencies, what normal form satisfies each of the relations: Users, Products, and Purchases?
- c) If any of the relations are not in 3NF/BCNF, then follow the methodology presented in the lecture to decompose them into a set of 3NF/BCNF relations that preserve the dependencies. Is there any opportunity to simplify your result?

2. Normalization

Suppose you are given a relation R with four attributes ABCD. For each of the following sets of functional dependencies, assuming that those are the only dependencies that hold for R, do the following:

- a) Identify all candidate keys for R.
- b) Identify the best normal form that R satisfies (1NF, 2NF, 3NF, BCNF).
- c) If R is not in 3NF/BCNF, then follow the methodology presented in the lecture to decompose it into a set of 3NF/BCNF relations that preserve the dependencies.

Functional dependencies:

1. $C \rightarrow D, C \rightarrow A, B \rightarrow C$
2. $B \rightarrow C, D \rightarrow A$
3. $ABC \rightarrow D, D \rightarrow A$
4. $A \rightarrow B, BC \rightarrow D, A \rightarrow C$
5. $AB \rightarrow C, AB \rightarrow D, C \rightarrow A, D \rightarrow B$

Note: This exercise is adapted from the textbook of Ramakrishnan and Gehrke.

